Treasure Ani-Joseph

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SUMMARY

Third-year Computer Science student and Data Analyst specializing in end-to-end data workflows: data simulation, SQL database architecture, and dashboard visualization. Strong focus on data integrity, utilizing out-of-sample testing, rigorous validation checks, and transparent limitation reporting. Complemented by full-stack engineering experience, bridging the gap between raw data infrastructure and production applications.

DATA & ANALYTICS PROJECTS

HELIOS | Energy-Management Simulation Model | Python, pandas, scikit-learn, NASA POWER API

- Built an hourly, 365-day energy simulation for a Nigerian household modeling grid, solar, battery, and generator usage, implementing strict per-hour energy-balance constraints to ensure data validity.
- Extracted and cleaned real NASA POWER climate data, identifying and correcting a hidden UTC offset anomaly by analyzing peak solar output hours against synthetic demand models.
- Evaluated 6 energy-dispatch strategies across 4 outage scenarios and 30 randomized outage years. Validated the optimal strategy on 20 unseen years to prevent overfitting.
- Discovered that maintaining a 50% battery reserve from 15:00 to 23:00 reduced operating costs by ~10% and generator usage by up to 33%.
- Proved and documented that this rules-based approach outperformed a logistic-regression outage predictor trained on separate historical data.

Project Concord | Unified Enterprise Data Platform | PostgreSQL, Supabase, SQL, BigQuery, Power BI

- Spearheaded database engineering for a 6-person team, architecting a hub-and-spoke PostgreSQL schema to centralize data across five business divisions.
- Engineered complex SQL views to resolve cross-divisional business queries, including supply deficit early-warnings, farmer yield-history for credit lending, and same-day consolidated revenue tracking.
- Designed and deployed Row Level Security (RLS) policies per division, ensuring data governance compliance with the Nigerian Data Protection Act.
- Pipelined structured database views into 4 interactive Power BI dashboards, enabling stakeholders to track KPIs while clearly documenting synthetic data limitations.

ADDITIONAL ENGINEERING EXPERIENCE

- Kablux (Contract Mobile Engineer): Engineered the real-time spatial data tracking and WebSocket messaging layer for a live ride-sharing and courier application (App Store & Google Play).
- LasuLearn (Mobile Developer): Shipped a companion e-learning application in React Native, implementing reliable real-time chat data flows and secure OTP authentication.

TECHNICAL SKILLS

- Data Analytics & Science: Python (pandas, NumPy, scikit-learn, Matplotlib), SQL, BigQuery, Jupyter, Data Modeling, ETL Pipelines.
- Databases & Visualization: PostgreSQL, Supabase, Power BI.
- Software Development: Linux, Git, React, TypeScript, React Native (Expo), Next.js, Node.js.

EDUCATION

B.Sc. Computer Science

University of Uyo, Akwa Ibom, Nigeria | Expected 2027

- Coursework highlights: System Design, Logic Programming, Software Engineering